

LAPORAN PRAKTIKUM

SISTEM OPERASI

PERTEMUAN 10

(Shell)



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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Jurnal

1. [40 Poin] Akan dimodifikasi shell dengan modifikasi syscall bernama **chname** yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.

Perhatikan sekarang syscall **chname** mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

- a. Pada prototypes.h chname diubah menjadi:

```
/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);
```

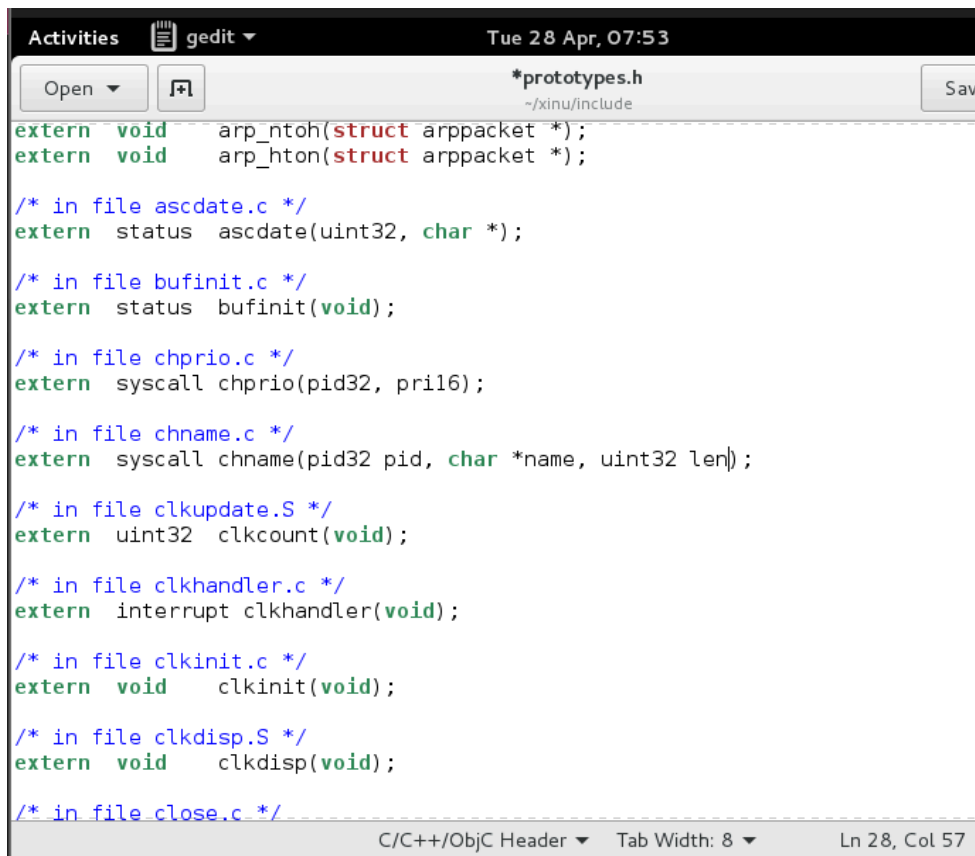
- b. Pada chname.c fungsi diubah dari:

```
pri16 chname(
    pid32      pid,          /* ID of process to change */
    pri16      newprio      /* New priority */
)
```

menjadi

```
pri16 chname(
    pid32      pid,          /* ID of process to change */
    char       *newname,
    uint32     namelen
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil.



```
Activities gedit Tue 28 Apr, 07:53
*prototypes.h
~/xinu/include
Open [file icon] Sav

extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdatetime.c */
extern status ascdatetime(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern syscall chname(pid32 pid, char *name, uint32 len);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */

C/C++/ObjC Header Tab Width: 8 Ln 28, Col 57
```

Activities gedit Tue 28 Apr, 08:03

Open *xsh_uptime.c ~/xinu/shell Save

```
/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");
chname(getpid(), "uptime", 6);
chprio(5,33);
kprintf("System uptime...\n");
return 0;
}
```

C Tab Width: 8 Ln 76, Col 37

Activities gedit Tue 28 Apr, 08:44

Open *chname.c ~/xinu/system Save

```
syscall chname(pid32 pid, char *name, uint32 len)
{
    intmask mask;
    struct procent *prptr;
    uint32 i;

    mask = disable();

    if (isbadpid(pid)){
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    if(name == NULL){
        restore(mask);
        return SYSERR;
    }
    for (i = 0; i < len && i < PNMLEN-1; i++){
        if (i >= len || name[i] == '\0'){
            break;
        }
        prptr->prname[i] = name[i];
    }
    prptr->prname[i] = '\0';
    restore(mask);
    return OK;
}
```

C Tab Width: 8 Ln 28, Col 18

```
/* in file xsh_ps.c */
extern shellcmd xsh_ps (int32, char *[]);

/* in file xsh_sleep.c */
extern shellcmd xsh_sleep (int32, char *[]);

/* in file xsh_udpdump.c */
extern shellcmd xsh_udpdump (int32, char *[]);

/* in file xsh_udpecho.c */
extern shellcmd xsh_udpecho (int32, char *[]);

/* in file xsh_udpeserver.c */
extern shellcmd xsh_udpeserver (int32, char *[]);

/* in file xsh_uptime.c */
extern shellcmd xsh_uptime (int32, char *[]);

/* in file xsh_help.c */
extern shellcmd xsh_help (int32, char *[]);

/* in file xsh_mycmd.c */
extern shellcmd xsh_mycmd (int32, char *[]);

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);

/* in file xsh_tee.c */
```

```
{"argecho", xsh_argecho},
{"arp", xsh_arp},
{"cat", xsh_cat},
{"clear", xsh_clear},
{"date", xsh_date},
{"devdump", xsh_devdump},
{"echo", xsh_echo},
{"help", xsh_help},
{"ls", xsh_ls},
{"kill", xsh_kill},
{"memdump", xsh_memdump},
{"memstat", xsh_memstat},
{"ns", xsh_ns},
{"netinfo", xsh_netinfo},
{"ping", xsh_ping},
{"ps", xsh_ps},
{"sleep", xsh_sleep},
{"tee", xsh_tee},
{"udp", xsh_udpdump},
{"udpecho", xsh_udpecho},
{"udpeserver", xsh_udpeserver},
{"uptime", xsh_uptime},
{"mycmd", xsh_mycmd},
{"namecmd", FALSE, xsh_namecmd},
{"?", xsh_help}

};
```

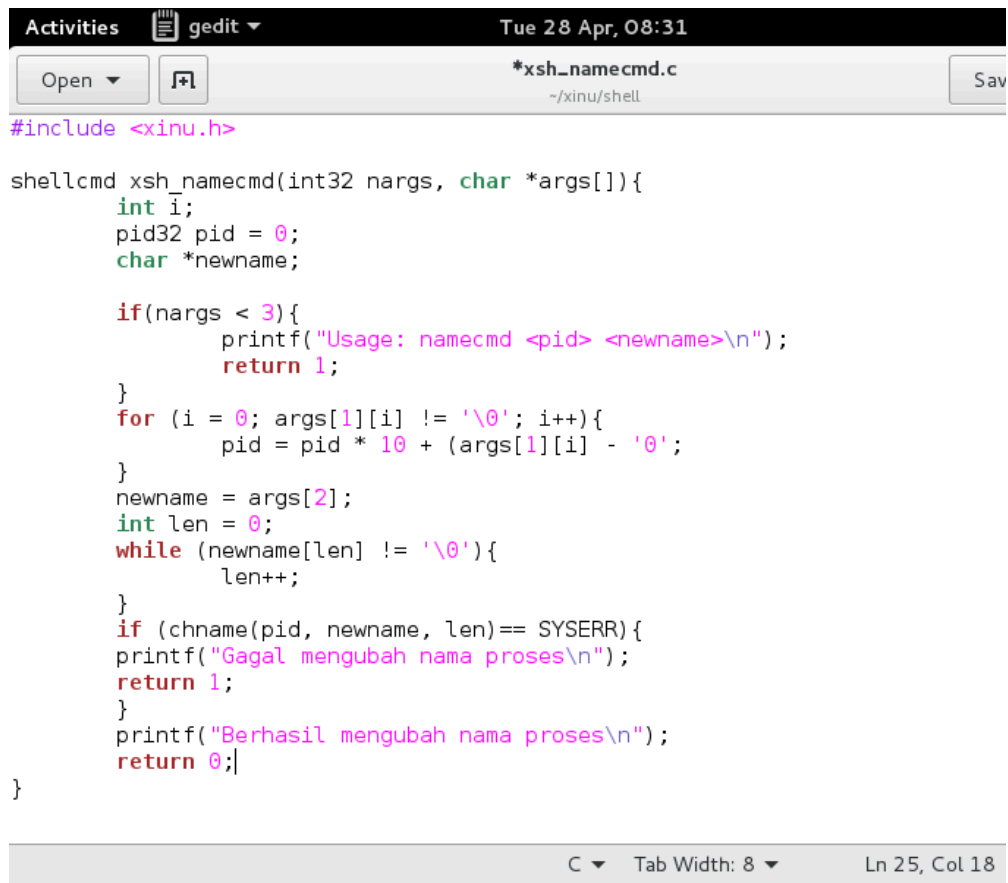
2. [40 Poin] Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

Berikut adalah kode dalam perintah baru namecmd:

```
printf("My new command");

// pid 3 = netin; mengubah nama proses pid 3 yaitu netin menjadi hello
chname(3,"hello",5);

return 0;
```



```
#include <xinu.h>

shellcmd xsh_namecmd(int32 nargs, char *args[]){
    int i;
    pid32 pid = 0;
    char *newname;

    if(nargs < 3){
        printf("Usage: namecmd <pid> <newname>\n");
        return 1;
    }
    for (i = 0; args[1][i] != '\0'; i++){
        pid = pid * 10 + (args[1][i] - '0');
    }
    newname = args[2];
    int len = 0;
    while (newname[len] != '\0'){
        len++;
    }
    if (chname(pid, newname, len) == SYSERR){
        printf("Gagal mengubah nama proses\n");
        return 1;
    }
    printf("Berhasil mengubah nama proses\n");
    return 0;
}
```

3. [20 Poin] Test hasilnya:
- Masuk ke terminal xinu
 - Jalankan perintah ps
 - Jalankan perintah namecmd
 - Jalankan perintah ps
 - Lihat nama proses telah berubah

```
Activities Terminal Tue 28 Apr, 08:21
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
Pid Name State Prio Ppid Stack Base Stack Ptr Stack Size Msg C
-----
0 prnull ready 0 0 0x005FDFFC 0x00FFFE4 8192 1
1 ipout wait 500 0 0x005FBFFC 0x005FBF30 8192 0
2 netin wait 500 0 0x005F9FFC 0x005F9ED0 8192 0
4 Main process recv 20 3 0x005E7FFC 0x005E7F34 65536 0
5 shell recv 50 4 0x005D7FFC 0x005D79AC 8192 0
7 ps curr 20 5 0x005F7FFC 0x005F7FC4 8192 0

xsh $ namecmd 5 baru
TRAP
Xinu trap!
exception 6 (invalid opcode) currrpid 8 (namecmd)
eflags 10282 eip 3 register dump:
eax 00000000 (0)
ecx 00000000 (0)
edx 00000000 (0)
ebx 00000000 (0)
esp 005F7FDC (6258652)
ebp 005F7FDC (6258652)
esi 00000000 (0)
edi 00000000 (0)

panic: Trap processing complete...

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | t
```